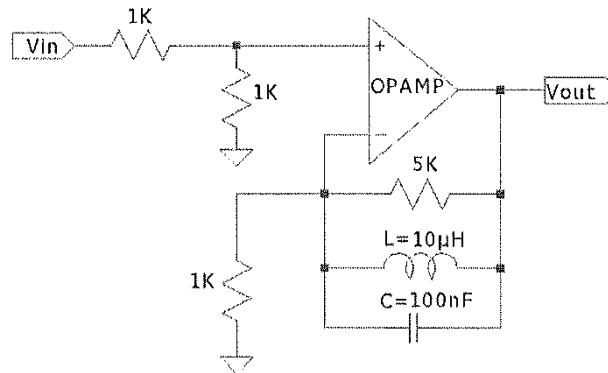


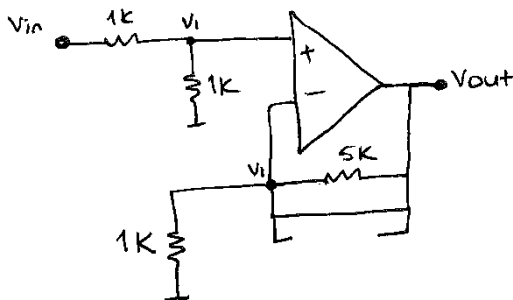
Question 1 [25 Points]:



For the circuit above, assume that the OPAMP is not saturated.

(a) [10 points] Find V_{out} if $V_{in} = 2$ Volts (DC).

For DC inputs, L is short circuit, C is open circuit



$$V_1 = V_{in} \cdot \frac{1K}{1K+1K} = \frac{V_{in}}{2}$$

$V_1 = V_{out}$ (because of short circuit)

So,

$$V_{out} = \frac{V_{in}}{2} = 1V$$

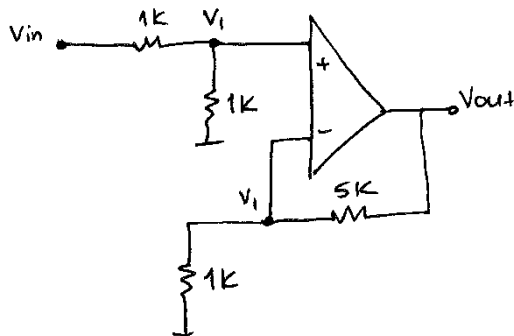
(b) [5 points] What is the resonance frequency (f_0) of the LC circuit in units of Hz?

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10 \times 10^{-6} \times 100 \times 10^{-9}}} = \frac{10^6}{2\pi}$$

$$f_0 = 159.15 \text{ kHz}$$

(c) [10 points] Find V_{out} if $V_{in} = 2 \cos(\omega_0 t + \pi/6)$ Volts, with ω_0 at resonance frequency.

At resonance, L & C cancel each other out.



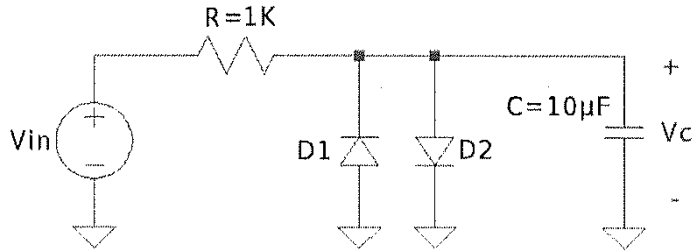
$$V_1 = V_{in} \cdot \frac{1K}{1K+1K} = \frac{V_{in}}{2}$$

$$V_{out} \cdot \frac{1K}{1K+5K} = V_1$$

$$\text{So, } V_1 = \frac{V_{in}}{2} = \frac{V_{out}}{6} \Rightarrow V_{out} = 3V_{in}$$

$$V_{out} = 3V_{in} = 6 \cos(\omega_0 t + \pi/6)$$

Question 2 [25 points]:



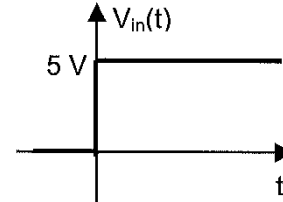
In the circuit above, the threshold voltage $V_{ON} = 0.6V$ for the diodes D1 and D2, and $V_C(0) = 0V$.

(a) [12 points] Find and plot $V_{out}(t)$ for $t > 0$ if $V_{in}(t)$ is as shown on the right.

$$\tau = RC = 1K \times 10\mu F = 10ms$$

* First, assume diodes are not there.

$$\left. \begin{array}{l} V_{init} = 0 \\ V_{final} = 5V \end{array} \right\} V_{out}(t) = 5 - 5e^{-t/10}$$



* D2 will turn ON when $V_{out}(t) \geq 0.6V$. Then, $V_{out} = 0.6V$ after that. Capacitor stops charging.

$$0.6 = 5 - 5e^{-t/10} \Rightarrow t_1 \approx 1.28ms$$

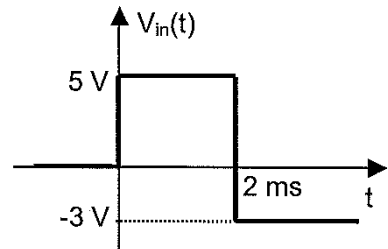
$$So, V_{out}(t) = \begin{cases} 5 - 5e^{-t/10} \text{ volts,} & 0 < t < 1.28ms \\ 0.6V & , t \geq 1.28ms \end{cases}$$

(b) [13 points] Find and plot $V_{out}(t)$ for $t > 0$ if $V_{in}(t)$ is as shown on the right.

Continue from part (a). τ is the same.

D2 turns OFF after $t > 2ms$

* at $t = 2ms$, $V_{out} = 0.6V = V_{init}$
 $V_{final} = -3V$



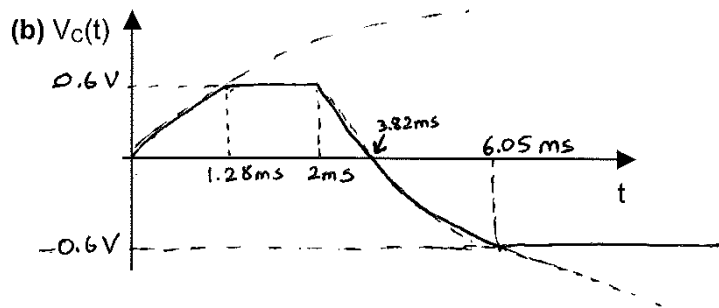
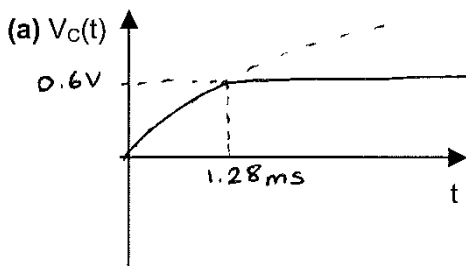
$$V_{out}(t) = -3 + (0.6 - (-3))e^{-(t-2)/10} = -3 + 3.6e^{-(t-2)/10}$$

* D1 will turn ON when $V_{out}(t) \leq -0.6V$. Then, $V_{out} = -0.6V$ after that. Capacitor stops charging.

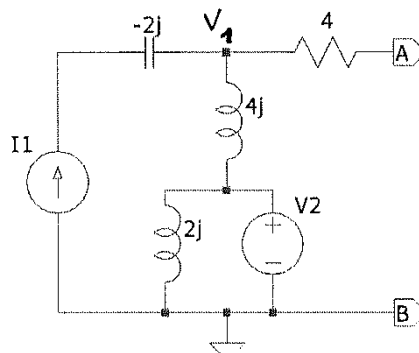
$$-0.6 = -3 + 3.6e^{-(t_2-2)/10} \Rightarrow t_2 \approx 6.05ms$$

$$V_{out}(t) = \begin{cases} 5 - 5e^{-t/10} \text{ volts} & , 0 < t < 1.28ms \\ 0.6V & , 1.28ms \leq t < 2ms \\ -3 + 3.6e^{-(t-2)/10} \text{ V} & , 2ms \leq t < 6.05ms \\ -0.6V & , t > 6.05ms \end{cases}$$

For (a) and (b), plot $V_{out}(t)$ below. Mark all important voltage levels and time points.



Question 3 [25 points]:



In the circuit above, $I_1(t) = 2 \sin(\omega t)$, $V_2(t) = 12 \cos(\omega t)$, and all of the impedances are provided in units of Ohms at frequency ω .

(a) [8 points] Find the Thevenin voltage $V_{TH}(t)$ in time domain, across nodes A&B.

$$I_1(t) = 2 \sin(\omega t) = 2 \cos(\omega t - \frac{\pi}{2}) \Rightarrow I_1 = 2 e^{-j\frac{\pi}{2}} = -2j$$

$$V_2(t) = 12 \cos(\omega t) \Rightarrow V_2 = 12 \quad \leftarrow \text{phasor notation} \rightarrow$$

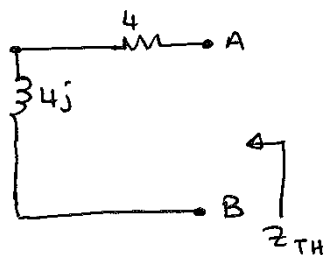
KCL at node 1:

$$-I_1 + \frac{V_1 - V_2}{4j} = 0$$

$$V_{TH} = V_1 = 4j I_1 + V_2 = 4j(-2j) + 12 = 20 \Rightarrow \boxed{V_{TH}(t) = 20 \cos(\omega t)} \text{ Volts}$$

(b) [9 points] Find the Thevenin equivalent impedance (Z_{TH}) across nodes A&B at frequency ω .

Kill all sources: Current source $\rightarrow$ open circuit
Voltage source $\rightarrow$ short circuit



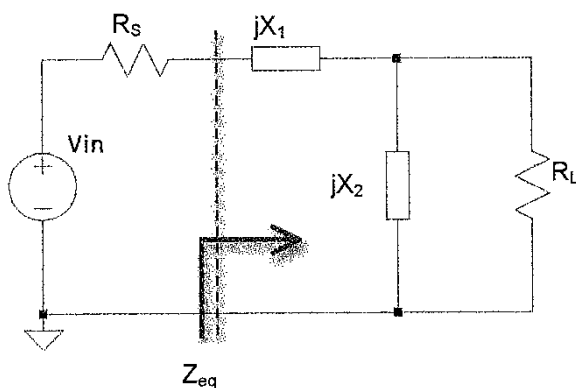
$$\boxed{Z_{TH} = 4 + 4j} \Omega$$

(c) [8 points] Find the Norton current $I_N(t)$ in time domain, across nodes A&B.

$$I_N = \frac{V_{TH}}{Z_{TH}} = \frac{20}{4 + 4j} = \frac{20}{4\sqrt{2} \angle \frac{\pi}{4}} = \frac{5}{\sqrt{2}} \angle -\frac{\pi}{4} \approx 3.54 \angle -\frac{\pi}{4}$$

$$\boxed{I_N(t) = 3.54 \cos(\omega t - \frac{\pi}{4})} \text{ Amps}$$

Question 4 [25 points]:



- (a) [10 points] For the circuit shown above, find Z_{eq} in terms of X_1 , X_2 , and R_L .

$$Z_{eq} = jX_1 + (jX_2 \parallel R_L)$$

$$= jX_1 + \frac{jX_2 R_L}{jX_2 + R_L} = \frac{jR_L(X_1 + X_2) - X_1 X_2}{jX_2 + R_L}$$

- (b) [10 points] Assuming $R_S = 20\Omega$ and $R_L = 100\Omega$, find the values of X_1 and X_2 that maximize the power dissipated on R_L .

For maximum power on R_L , we need $Z_{eq} = R_S$

$$Z_{eq} = \frac{jR_L(X_1 + X_2) - X_1 X_2}{jX_2 + R_L} = R_S$$

$$jR_L(X_1 + X_2) - X_1 X_2 = jX_2 R_S + R_L R_S$$

* Real and imaginary parts must match on both sides of this equation.

$$R_L(X_1 + X_2) = X_2 R_S \quad \text{and} \quad -X_1 X_2 = R_L R_S$$

$$\Downarrow$$

$$100(X_1 + X_2) = 20X_2$$

$$X_2 = -\frac{5}{4}X_1$$

$$\Downarrow$$

$$X_1 X_2 = -2000$$

$$X_1 \left(-\frac{5}{4}X_1\right) = -2000$$

$$X_1^2 = 1600$$

$$\boxed{X_1 = 40\Omega}$$

$$\boxed{X_2 = -50\Omega}$$

Alternative solution:

$$\boxed{X_1 = -40\Omega}$$

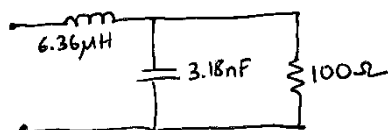
$$\boxed{X_2 = 50\Omega}$$

- (c) [5 points] Find the values of inductance and/or capacitance values that will yield the X_1 and X_2 obtained in part (b). Assume that the circuit operates at $f = 1\text{ MHz}$.

$$X_1 = 40\Omega \quad \text{and} \quad X_2 = -50\Omega :$$

$$j40 = j\omega L \Rightarrow L = \frac{40}{2\pi \cdot 10^6} = 6.36\mu\text{H}$$

$$-j50 = \frac{1}{j\omega C} \Rightarrow C = \frac{1}{50 \cdot 2\pi \cdot 10^6} = 3.18\text{nF}$$



Alternative solution:

$$X_1 = -40\Omega \quad \text{and} \quad X_2 = 50\Omega$$

$$-j40 = \frac{1}{j\omega C} \Rightarrow C = \frac{1}{2\pi \cdot 10^6 \cdot 40} = 3.98\text{nF}$$

$$j50 = j\omega L \Rightarrow L = \frac{50}{2\pi \cdot 10^6} = 7.95\mu\text{H}$$

